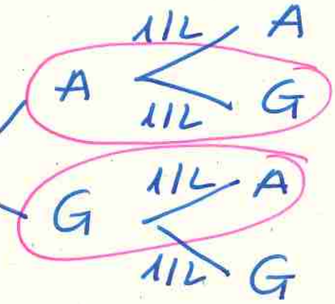
ADIBIDEA



A: aurpegia
G: gurutza.

p: aurpegia ateratzea
q: gurutza ateratzea (aurpegia ez ateratzea).

$$P(\text{aurpegi bat}) = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}+\frac{1}{2} = \frac{2}{4} = \frac{1}{2}$$

rmulagaz kalkulua.

$X \sim B\left(2, \frac{1}{2}\right)$

$$P(X=k) = \binom{n}{k} p^k q^{n-k}$$

$$P(X=1) = \binom{2}{1}\left(\frac{1}{2}\right)^1\left(\frac{1}{2}\right)^1 = 2\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{2}$$

ADIBIDEA

o aldiz bota $\longrightarrow n=10$.

gertaera: Arrakasta (6 zenbakia) $p=1/6$; Ez arrakasta (6 ez) $q=5/6$.

$X \sim B(10, 1/6)$

6 zenbakia bi aldiz ateratzea.

$$P(X=2) = \binom{10}{2}\cdot(1/6)^2\cdot(5/6)^8 = 45\cdot(1/6)^2\cdot(5/6)^8 = 0{,}29$$

BOST aldiz baino gehiago ateratzearen prob.

$$P(X>5) = P(X=6)+P(X=7)+P(X=8)+P(X=9)+P(X=10).$$

419. orrialdetik

1/ $B(10; 0{,}4) \longrightarrow n=10 \quad p=0{,}4. \quad q=0{,}6$

$P(x=0) = \binom{10}{0} 0{,}40^{0} \cdot 0{,}6^{10-0} = 1 \cdot 1 \cdot 0{,}6^{10} = \underline{\underline{0{,}00605}}$

$P(x=3) = \binom{10}{3} 0{,}4^{3} \cdot 0{,}6^{10-3} = \underline{\underline{0{,}215}}$

$P(x=5) = \binom{10}{5} 0{,}4^{5} \cdot 0{,}6^{10-5} = \underline{\underline{0{,}2007}}$

$P(x=10) = \binom{10}{10} 0{,}40^{10} \; 0{,}6^{10-0} = 1 \cdot 0{,}4^{10} \cdot 1 = \underline{\underline{0{,}000105}}$

$\mu = np = 10 \cdot 0{,}4 = \underline{\underline{4}}.$

$\sigma = \sqrt{npq} = \sqrt{10 \cdot 0{,}4 \cdot 0{,}6} = \underline{\underline{1{,}55}}. \longrightarrow \underline{\underline{N(4; 1{,}55)}}.$

2/. $x \sim B(n,p)$

$n = 7$

$p = 1/2.$ (arrokasta: aurpegia ateratzea)

$q = 1/2$ (ez arrokasta: gurutza ateratzea)

$P(x=3) = \binom{7}{3} \cdot 0{,}5^{3} \cdot 0{,}5^{7-3} = \underline{\underline{0{,}2734}}$

$P(x=5) = \binom{7}{5} \; 0{,}5^{5} \cdot 0{,}5^{7-5} = \underline{\underline{0{,}1641}}$

$P(x=6) = \binom{7}{6} \; 0{,}5^{6} \cdot 0{,}5^{7-6} = \underline{\underline{0{,}0547}}.$

$\mu = 7 \cdot 0{,}5 = \underline{\underline{3{,}5}}$

$\sigma = \sqrt{7 \cdot 0{,}5 \cdot 0{,}5} = \underline{\underline{1{,}32}}$

$\mu = np$

$\sigma = \sqrt{npq}$

$x \sim B(7, 1/2) \longrightarrow N(3{,}5; 1{,}32)$

Banak. binomiala (diskretua) — Banaketa normala.

3. ADIBIDEA

4. ADIBIDEA

Banaketa binomiala.

X = ondo erantzundako galdera kopurua.

$n = 20 \rightarrow \boxed{B(20, 1/4)}$

$p = 1/4$

$q = 3/4$

Batezbestekoa $\mu = n \cdot p = 20 \cdot 1/4 = 5.$ $\boxed{\mu = 5}$

Desbiderazte tipikoa $\sigma = \sqrt{npq} = \sqrt{20 \cdot 1/4 \cdot 3/4} = \boxed{1.94}$

5. ADIBIDEA

Banaketa binomiala da:

$n = 8$ saiakera egiten dauz. eta bakarrik "arrakasta" edo "arrakasta eza" kontsideratzen doguz, prob konstanteak izanik (penaltia sartzeko %50 prob) eta froga bakoitzaren emaitza aurrekoetatik independ.

Definitzen dugu x eta banaketa:

x = sartutako penalti kopurua.

$n = 8 \longrightarrow \boxed{X \approx B(8; 0,9)}$

$p = 0,9$

$q = 0,1$

a) <u>7 penalti sartzeko prob.</u> $P(x=7) = \binom{8}{7} \cdot 0,9^7 \cdot 0,1^{8-7} = \underline{\underline{0,3826}}$

7 penalti sartzeko probabilitatea 0,3826-koa da

% 38.

b) <u>8 penalti sartzeko prob</u> $P(x=8) = \binom{8}{8} \cdot 0,9^8 \cdot 0,1^0 = \underline{\underline{0,4305}}$

c) <u>6 penalti baino gehiago sartzeko prob</u> ($k > 6$.)

$P(x>6) = P(x=7) + P(x=8) = 0,3826 + 0,4305 =$

$= \underline{\underline{0,8131}}$

d) <u>gutxienez 6 penalti sartzeko prob.</u> ($k \geq 6$).

$P(x \geq 6) = P(x=6) + P(x=7) + P(x=8)$

$P(x=6) = \binom{8}{6} 0,9^6 \cdot 0,1^{8-6} = 0,1488$

$P(x \geq 6) = 0,1488 + 0,3826 + 0,4305 = \underline{\underline{0,9619}}$

ADIBIDEA 6. X = tiroa asmatzea. J.

$n = 10$ (10 aldiz tiro egin)

$q = 0{,}1$ (arrakasta)

$p = 0{,}9$ (arrakasta ez)

$\Rightarrow \boxed{X \sim B(10; 0{,}1)}$

Banaketa binomiala.

a) Behin ere EZ asmatzea ($K=0$).

$$P(X=0) = \binom{10}{0} \cdot 0{,}1^{0} \cdot 0{,}9^{10-0} = \underline{\underline{0{,}3487}}$$

b) Behin bakarrik asmatzea. ($K=1$).

$$P(X=1) = \binom{10}{1} \cdot 0{,}1^{1} \cdot 0{,}9^{10-1} = \underline{\underline{0{,}3874}}.$$

c) Bi baino gutxiagotan asmatzea ($K<2$).

$$P(X<2) = P(X=0) + P(X=1) = 0{,}3487 + 0{,}3874 = \underline{\underline{0{,}7361}}$$

d) Gehienez bi aldiz $K \leq 2$.

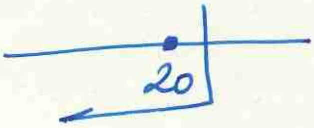
$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$P(X=2) = \binom{10}{2} \, 0{,}1^{2} \cdot 0{,}9^{10-2} = 0{,}1937$$

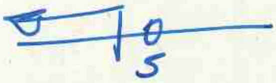
$$P(X \leq 2) = 0{,}3487 + 0{,}3874 + 0{,}1937 = \underline{\underline{0{,}9298}}$$

YATES-en ZUZENKETA

$P(x \le 20) = P(x' \le 20{,}5)$ $\qquad P(x \le k) = P(x' \le (k+0{,}5))$

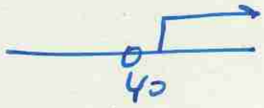


$P(x < 5) = P(x' < 4{,}5)$ $\qquad P(x < k) = P(x' < k-0{,}5)$

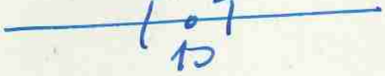


$P(x \geq 4) = P(x' \geq 3{,}5)$ $\qquad P(x \geq k) = P(x' \geq k-0{,}5)$

$P(x > 40) = P(x' > 40{,}5)$ $\qquad P(x > k) = P(x' > k+0{,}5)$



$P(x = 10) =$ $\qquad P(x = k) = P(k-0{,}5 \le x' \le k+0{,}5)$



$P(9{,}5 \le x' \le 10{,}5)$

16. Dado bat 1000 aldiz botatzen dogu. Zer probabilitate dago bostekoen kopurua 100 baino txikiagoa izateko?

Banaketa binomiala $B(n,p) \rightarrow \boxed{X \sim B(1000, \frac{1}{6})}$

Banaketa binomiala, banaketa normala erabiltzeko bihurketa egiteko baldintzak:

$n \geq 10$	$n = 1000$
$np \geq 5$	$np = 1000 \cdot \frac{1}{6} = 166,67 \geq 5$
$nq \geq 5$	$nq = 1000 \cdot \frac{5}{6} = 833,33 \geq 5$

Banaketa binomiala, normalera hurbiltzeko zuzenketa gin behar da, Yatesen zuzenketarekin.

B. BINOMIALA	B. NORMALA	B. NORM. TIPIFIKATUA
$X \sim B(n,p)$ $\Rightarrow$	$X' \sim N(np, \sqrt{npq})$ $\rightarrow$	$Z \sim N(0,1)$
Yatesen zuzenketa	$N(\mu, \sigma)$, $\mu = np$, $\sigma = \sqrt{npq}$	TIPIFIKATU: $z = \frac{x-\mu}{\sigma}$

$$X \sim B(1000, \tfrac{1}{6}) \rightarrow X' \sim N(166,67; 11,79) \rightarrow Z \sim N(0,1)$$

$$P(X < 100) = P(X' < 99,5) = P\left(Z < \frac{99,5 - 166,67}{11,79}\right)$$

$$= P(Z \leq -5,7) = P(Z \geq 5,7) = 1 - P(Z \leq 5,7) =$$

$$= 1 - 1 = \underline{\underline{0}}$$

$\boxed{17}$ Txanpon bat 400 bider bota.

a) Lortutako aurpegien kopurua > 200

Txanpon bat 400 bider botatzearen prob. jarraitzen dau BANAKETA BINOMIALA $\boxed{B(400, 1/2)}$

Banak. binomiala, ban. normalera hurbiltzeko baldintzak:

$n \geq 100 \qquad n = 400$

$np \geq 5 \qquad n \cdot p = 400 \cdot 1/2 = 200 \geq 5 \qquad \sqrt{npq} = \sqrt{400 \cdot \frac{1}{2} \cdot \frac{1}{2}} = 10.$

$nq \geq 5 \qquad nq = 400 \cdot 1/2 = 200 \geq 5$

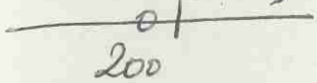
Banaketa binomiala, normalera hurbiltzeko, Yatesen zuzenketa egin behar da:

BINOMIALA $\longrightarrow$ B. NORMALA $\longrightarrow$ B. NORMAL TIPIFIK.

$$X \sim B(n,p) \xrightarrow{\text{Yates}} X' \sim N(\underbrace{np}_{\mu}, \underbrace{\sqrt{npq}}_{\sigma}) \xrightarrow{\text{Tipifikatu}} Z \sim N(0,1) \qquad Z = \frac{X-\mu}{\sigma}$$

$$X \sim B(400, 1/2) \rightarrow X' \sim N(200, 10) \rightarrow Z' \sim N(0,1)$$

$$P(X > 200) = P(X' \geq 200{,}5) = P\left(Z \geq \frac{200{,}5 - 200}{10}\right) =$$



$$= P(Z \geq 0{,}05) = 1 - P(Z \leq 0{,}05) = 1 - 0{,}5199 =$$

$$= \underline{\underline{0{,}4801}}$$

b) 180 eta 220 artean egoteko.

Binomiala $\longrightarrow$ (Yates) $\longrightarrow$ Normal.

$$P(180 < X < 220) = P(180{,}5 \leq X \leq 219{,}5) =$$

$$= P\left(\frac{180{,}5-200}{10} \leq Z \leq \frac{219{,}5-200}{10}\right) = P(-1{,}95 \leq Z \leq 1{,}95) =$$

$$= P(Z \leq 1{,}95) - P(Z \leq -1{,}95) = P(Z \leq 1{,}95) - P(Z \geq 1{,}95) =$$

$$= P(Z < 1{,}95) - (1 - P(Z \leq 1{,}95)) = 2\,P(Z \leq 1{,}95) - 1 =$$

$$= 2\;0{,}9744 - 1 = \boxed{0{,}9488}$$

40

$n = 2500$ $\quad p = 0{,}46 \longrightarrow q = 0{,}54$

Banaketa binomiala $\quad B(2500;\ 0{,}46)$

$n \geq 10$ $\quad n = 2500 \geq 10$

$np \geq 5$ $\quad np = 2500 \cdot 0{,}46 = 1150 \geq 5$ $\quad \Longrightarrow$ Banaketa normalera hurbilketa egin daiteke

$nq \geq 5$ $\quad nq = 2500 \cdot 0{,}54 = 1350 \geq 5$

$X \sim B(n,p) \longrightarrow X' \sim N(np, \sqrt{npq})$ $\quad \sqrt{npq} = \sqrt{2500 \cdot 0{,}46 \cdot 0{,}54} = 24{,}92.$

B. binomiala $\quad$ B normala

$B(2500;\ 0{,}46)$ $\quad N(1150;\ 24{,}92)$

BINOMIALA $\xrightarrow{\text{YATES}}$ NORMALA $X' \sim N(1150;\ 24{,}92)$

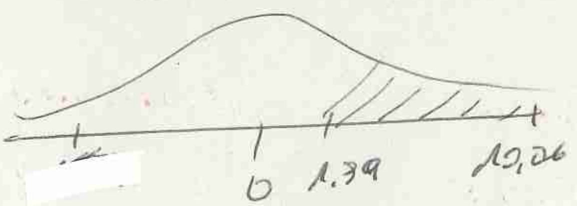
$P(1200 \leq X \leq 1400) = P(1199{,}5 \leq X' \leq 1400{,}5) =$

Normal tipifikatua $\quad Z \sim N(0,1)$

$= P\left(\frac{1199{,}5 - 1150}{24{,}9} \leq Z \leq \frac{1400{,}5 - 1150}{24{,}9}\right)$

$= P(1{,}99 \leq Z \leq 10{,}06) = P(Z \leq 10{,}06) - P(Z \leq 1{,}99) =$

$= 1 - 0{,}9767 = \underline{\underline{0{,}0233}}$



41.

$n = 50$

$p = 1/3$ arrakasta

$q = 2/3$ arrakasta ez.

$\Big\}\; B(50;\, 1/3)$

GAINDITU $= 25$. $\quad p?$

OSO ONDO $= 35$ $\quad p?$

BIKAIN $= 45$. $\quad p?$

$n \geq 10 \qquad n = 50$

$np \geq 5 \qquad np = 50 \cdot 1/3 = 16{,}67 \Rightarrow$ Banaketa normalera hurbilketa egin daiteke.

$nq \geq 5 \qquad nq = 33{,}33$

Banaketa normala $N(np, \sqrt{npq})$

$\sigma = \sqrt{npq} = \sqrt{50 \cdot \frac{1}{3} \cdot \frac{2}{3}} = 3{,}33$

$\boxed{X' \sim N(16{,}67;\, 3{,}33)}$

Notak izango ziren:

GAINDITU. $[25, 35)$

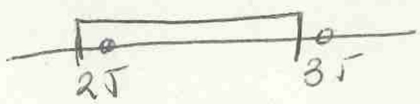
OSO ONDO $[35, 45)$

BIKAIN. $[45, 50]$

GAINDITU: BINOMIALA $\quad$ NORMALA $\quad$ NORMAL TIPIFIK

$X \sim B(50, 1/3) \quad X' \sim N(16{,}67;\, 3{,}33) \quad Z \sim N(0,1)$

$$P(25 \leq X < 35) \xrightarrow{\text{YATES}} = P(24{,}5 \leq X' < 34{,}5) = P\left(\frac{24{,}5 - 16{,}67}{3{,}33} \leq Z \leq \frac{34{,}5 - 16{,}67}{3{,}33}\right)$$



$$= P(2{,}35 \leq Z \leq 5{,}35) =$$

$$= 1 - 0{,}9906 = \boxed{0{,}0094}$$

41.(2) OSO ONDO

$$P(35 \leq x < 45) = P(34{,}5 \leq x' \leq 44{,}5) =$$

$$= P\left(\frac{34{,}5-16{,}67}{3{,}33} \leq z \leq \frac{44{,}5-16{,}67}{3{,}33}\right)$$

$$= P(5{,}35 \leq z \leq 8{,}36) =$$

$$= P(z \leq 8{,}36) - P(z \leq 5{,}35) =$$

$$= 1 - 1 = \underline{\underline{0}}$$

BIKAIN.

$$P(x = 45) = P(44{,}5 \leq x' \leq 45{,}5) =$$

$$= P\left(\frac{44{,}5-16{,}67}{3{,}33} \leq z \leq \frac{45{,}5-16{,}67}{3{,}33}\right)$$

$$= P(8{,}36 \leq z \leq 8{,}66) =$$

$$= P(z \leq 8{,}66) - P(z \leq 8{,}36) =$$

$$= 1 - 1 = \underline{\underline{0}}$$

42. $p = 0{,}1$. $n = 300$. } Banaketa binomiala jarraitzen dau

X = akatsdun tornilo kopurua

$\boxed{B(300;\, 0{,}1)}$

Banaketa binomiala, normalera hurbiltzeko ezaugarriak:

$n \geq 10$	$n = 300$	$\sigma = \sqrt{npq} = \sqrt{300 \cdot 0{,}1 \cdot 0{,}9}$
$np \geq 5$	$np = 300 \cdot 0{,}1 = 30 \geq 5$	$= 5{,}2$
$nq \geq 5$	$nq = 300 \cdot 0{,}9 = 270 \geq 5$	

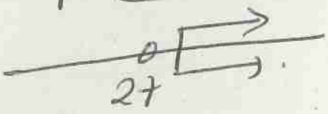
Binomiala, normalera hurbilketa egiteko Yatesen zuzenketa aplikatu behar da.

B. BINOMIALA	B. NORMALA	B. NORMALA TIPIKOA
$X \sim B(300; 0{,}1)$	$X' \sim N(30; 5{,}2)$	$Z \sim N(0,1)$
	$\mu = np$	
	$\sigma = \sqrt{npq} = 5{,}2$	

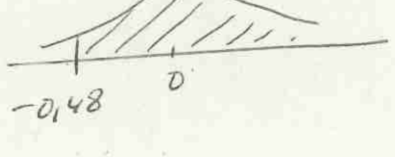
a) **Akastunak %9tik gora izateko.**

$\% 9 \cdot 300 \rightarrow 0{,}09 \cdot 300 = 27$.

$$P(X > 27) = P(X' \geq 27{,}5) = P\left(Z \geq \frac{27{,}5 - 30}{5{,}2}\right)$$



$$= P(Z \geq -0{,}48) = P(Z \leq 0{,}48) =$$



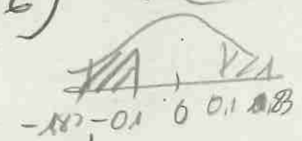
$$= \boxed{0{,}6844}$$

b) 20 eta 30 artikook $\quad B(300;0,1) \rightarrow N(30;5,2)$

$P(20 < x < 30) = P(20,5 < x' < 29,5) =$

$P\left(\frac{20,5-30}{5,2} < z < \frac{29,5-30}{5,2}\right) =$

$= P(-1,83 < z \; \underset{-0,1}{-0,096}) = P(z < -0,1) - P(z < -1,83) =$



$= P(z < 1,83) - P(z < 0,1) =$

$= 0,9664 - 0,5398 = \boxed{0,4266}$